

DEEPAKRAJ B

GENAI SOFTWARE ENGINEER | LANGGRAPH, MULTI-STEP WORKFLOWS & BACKEND SYSTEMS

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PROFILE

Generative AI and Backend Software Engineer with hands-on experience building production-grade LLM applications, stateful multi-step workflows, and scalable backend services. Proven track record designing and shipping an **NL2SQL Query Engine** using **LangGraph**, **FastAPI**, and **SQLGlot**, and architecting the **AutoSDLC PRD-to-Execution automation platform**. Skilled in context management, prompt engineering, structured JSON outputs, database schema grounding, and validation guardrails. Proficient in **Python**, **FastAPI**, **Redis**, **Docker**, and **AWS**, with a strong focus on safe LLM execution, deterministic parsing, and robust error recovery.

PROFESSIONAL EXPERIENCE

Software Engineer | Kritilabs Technologies Pvt. Ltd., Chennai, India July 2024 - Present

NL2SQL Query & Reasoning Engine

FastAPI, LangGraph, SQLGlot, Redis, PostgreSQL, MySQL, Docker, AWS

- Architected and shipped an **NL2SQL Query Engine** using **LangGraph** and **FastAPI**, converting complex natural language queries into verified SQL over enterprise MySQL and PostgreSQL databases.
- Implemented multi-step graph workflows featuring intent classification, dynamic schema introspection, automatic join path resolution, and semantic error-correction loops.
- Built AST-based validation and security guardrails using **SQLGlot** to block destructive DDL/DML operations (DROP, DELETE, UPDATE) and prevent SQL injection vulnerabilities.
- Integrated **Redis** for stateful conversational session tracking and schema metadata caching, reducing end-to-end translation latency by **35%**.
- Packaged backend microservices with **Docker** and deployed on **AWS (EC2, S3)** with automated CI/CD pipelines and structured JSON logging for observability.

AutoSDLC: PRD-to-Execution Automation Platform

Python, FastAPI, LangChain, Pydantic, Redmine REST API, Pytest

- Developed an end-to-end automation engine that parses complex Product Requirement Documents (PRDs) into structured Epics, User Stories, acceptance criteria, and actionable development tasks.
- Implemented **Pydantic-based JSON schema validation** and automated retry mechanisms to eliminate hallucinations and guarantee deterministic output structures.
- Integrated directly with **Redmine REST APIs** to automatically populate backlogs and task boards, reducing manual documentation and sprint planning time by **70%**.
- Designed modular prompt templates and evaluation test suites using **Pytest** to systematically benchmark parsing accuracy across diverse requirement formats.

Enterprise Backend Services & Edge Systems

Python, Java, Spring Boot, Frappe, MySQL, REST APIs

- Built custom backend modules and REST APIs for a 700-employee manufacturing platform, streamlining inventory tracking and shop-floor data collection.
- Assisted in backend integration for edge computer vision and IoT tracking deployments serving **Indian Railways**, **IOCL**, and military logistical operations.

TECHNICAL SKILLS

GenAI & LLM Frameworks: LangGraph, LangChain, FastMCP, Model Context Protocol, Multi-Step Reasoning

Backend & Languages: Python (FastAPI, AsyncIO), Java (Spring Boot), SQL, RESTful APIs, Webhooks

Cloud & DevOps: Docker, Docker Compose, AWS (EC2, S3), Linux / Bash, Git, CI/CD Pipelines

Validation & Guardrails: SQLGlot, Pydantic, Schema Grounding, AST Validation, Error Recovery

Databases & Caching: PostgreSQL, MySQL, Redis (State & Session Cache), ChromaDB (Vector Search)

Prompt & Evaluation: Context Window Optimization, Structured JSON Outputs, Pytest, Postman

EDUCATION

B.Tech in Computer Science & Engineering | SASTRA Deemed University, Thanjavur CGPA: 7.2955 | 2020 - 2024

CERTIFICATIONS

- Machine Learning Specialization - Simplilearn
- Python Crash Course - Google
- Software Engineering Specialization - IBM
- Deep Learning & Neural Networks Fundamentals